Combining the Neuman and Boulton Models for Flow to a Well in an Unconfined Aquifer

by Allen F. Moench[a]

Abstract

A Laplace transform solution is presented for flow to a well in a homogeneous, water-table aquifer with noninstantaneous drainage of water from the zone above the water table. The Boulton convolution integral is combined with Darcy's law and used as an upper boundary condition to replace the condition used by Neuman. Boulton's integral derives from the assumption that water drained from the unsaturated zone is released gradually in a manner that varies exponentially with time in response to a unit decline in hydraulic head, whereas the condition used by Newman assumes that the water is released instantaneously. The result is a solution that reduces to the solution obtained by Neuman as the rate of release of water from the zone above the water table increases. A dimensionless fitting parameter, γ, is introduced that incorporates vertical hydraulic conductivity, saturated thickness, specific yield, and an empirical constant α_1, similar to Boulton's α. Results show that theoretical drawdown in water-table piezometers is amplified by noninstantaneous drainage from the unsaturated zone to a greater extent than drawdown in piezometers located at depth in the saturated zone. This difference provides a basis for evaluating γ by type-curve matching in addition to the other dimensionless parameters. Analysis of drawdown in selected piezometers from the published results of two aquifer tests conducted in relatively homogeneous glacial outwash deposits but with significantly different hydraulic conductivities reveals improved comparison between the theoretical type curves and the hydraulic head measured in water-table piezometers.

Introduction

Analytical solutions for flow to a well in a water-table aquifer, developed by Neuman (1972, 1974), were derived under the assumption that the drainage of pores from the zone above the water table occurs instantaneously in response to a decline in the elevation of the water table. This common assumption has been criticized for not being realistic and running counter to experimental findings (see Nwankwor et al., 1992, and Narasimhan and Zhu, 1993). In spite of the possible limitations inherent in the assumption, the Neuman model has been used successfully by many hydrogeologists in the analysis of pumping tests conducted in water-table aquifers. One of the primary features of the model is that it allows drawdown to vary continuously in the vertical as well as the horizontal directions, thus retaining the full three-dimensional, axisymmetric character of the flow regime. Another feature is that the model accounts for aquifer compressibility.

The two-dimensional, axisymmetric model developed by Boulton (1954, 1963) is also used to analyze pumping-test data from water-table aquifers. Boulton's original model does not account for vertical components of flow in the aquifer and cannot be used for partially penetrating wells or for estimating vertical hydraulic conductivity, but it does take into consideration noninstantaneous release of water from the zone above a falling water table. Boulton (1954) assumed that the drainage of pores in this zone occurs as an exponential function of time in response to a step change in hydraulic head in the aquifer. Such a response can be derived theoretically for drainage at the base of a vertical column of incompressible soil within which the water table lies (see, for example, Dagan and Kroszynski, 1971, and Cooley and Case, 1973). Indeed, Boulton (1954) actually developed his theory under the assumption that the delayed drainage occurs through fine-grained material overlying a coarse-grained aquifer. He later argued (Boulton, 1973) that a thin stratum of low-permeability soil through which the water table is falling is a necessary condition for the slow drainage described by Boulton (1954, 1963). Boulton and Streltsova (1975) extended this work by considering an aquifer overlain by a water-table aquitard (or layer of low-permeability material containing the water table). Because the boundary condition used for water table in the aquitard is the same as that used by Neuman (1974) for the upper boundary of the aquifer, their solution reduces to Neuman's solution as the aquitard thickness becomes zero. An earlier treatment of the problem of flow to a well in an aquifer overlain by a water-table aquitard is given by Cooley and Case (1973).

Vachaud (1968) demonstrated by means of laboratory drainage experiments on river sand and Narasimhan and Zhu (1993) showed by numerical experiments that the exponential approximation is an oversimplification of the actual drainage process. Dagan and Kroszynski (1971), Kroszynski and Dagan (1975), and Cooley and Case (1973) provide alternative formulations which derive from solution of linearized forms of the Richards equation and from assumed approximate representations of the flow properties of the unsaturated zone. Such formulations should improve upon the simple exponential relation. Use of these or similar relatively complicated analytical representations of the

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drainage process is beyond the scope of the present paper. It is clear, however, that the inclusion of time-dependent drainage in any reasonable form would be an improvement over the assumption of instantaneous drainage.

Dagan (1967) developed a model that is similar to that of Neuman (1972, 1974) except that it does not allow for compressiblity of the aquifer. Dagan's work was extended by Boulton and Pontin (1971) to account for effects of delayed yield. This was accomplished by modifying the free-surface condition used by Dagan (1967) so that, in addition to an instantaneous release of water by vertical drainage from the zone above the water table, two other components that account for short-term and long-term delayed yield were added. In this context, however, it is not clear what is meant by "instantaneous yield," what its magnitude is, and at what point in time the short-term delayed yield takes over from the instantaneous yield. If the process being described is one involving delayed yield, the release of water from the unsaturated zone should be continuously changing in response to a step change in the elevation of the water table and there should be no instantaneous yield. In an appendix of another paper, Boulton (1970) presents a solution to a similar problem, involving a fully penetrating pumped well, for which effects of delayed yield were considered. The manner in which delayed yield was included in the development, however, is not clear as no derivation is provided. Also, the solution does not appear to have been used by Boulton for data analysis.

In support of Boulton's pioneering work the recent findings by Akindunni and Gillham (1992), Nwankwor et al. (1992), Narasimhan and Zhu (1993), and Moench (1994) demonstrate that effects of slow drainage from the zone above the water table are manifest in some piezometers, especially those located near the water table, during pumping tests in unconfined aquifers. Kroszynski and Dagan (1975) demonstrated from a theoretical viewpoint, by means of a numerical model and an approximate analytical model, that such effects would occur but that they could be neglected for purposes of pumping-test analysis. A similar conclusion was reached by Brutsaert and El-Kadi (1984) who presented a criterion, similar to that used by Kroszynski and Dagan (1975) in their analytical solution, for determining the conditions under which effects of flow from the unsaturated zone might be important in pumping-test analysis. The criterion states that if a representative thickness of the capillary fringe, Ψ_t, is much less than the saturated thickness of the aquifer, b, then effects of the unsaturated zone upon drawdown in the saturated zone can be neglected. The difficulty with this criterion is that the required magnitude of the ratio Ψ_t / b may depend upon the location of the particular piezometers being used. In two instances described by Moench (1994) effects of the unsaturated zone are negligible for deep-seated piezometers and for long-screened observation wells but not negligible for water-table piezometers. In one instance Ψ_t / b is approximately 0.05 and in the other it is probably much less than that. In the former instance, temporal and spatial measurements of water content and hydraulic head within the unsaturated zone in the course of a pumping test (Nwankwor et al., 1992) convincingly demonstrate the direct connection between delayed release of water from the unsaturated zone and observations of hydraulic head in piezometers located at the water table.

The purpose of this paper is to incorporate the Boulton (1954) convolution integral for delayed drainage into the Neuman (1972, 1974) model and to use the resulting solution to improve the match between theoretical type curves and drawdown measured in water-table piezometers during pumping tests. An empirical constant, α_1, similar to Boulton's α, is introduced. The approach is similar to that of Boulton and Pontin (1971) but eliminates the ambiguous "instantaneous yield" term from consideration. It also allows for aquifer compressibility. The solution is mathematically similar to that presented by Boulton and Streltsova (1975) but is conceptually different in that no stratum of low-permeability material is assumed to overlie the aquifer. No attempt is made to quantitatively relate the empirical constant to physically defined parameters in the zone above or near the water table. Assumptions required to make such an association are deemed unjustified. The solution presented in this paper is obtained in Laplace space and inverted numerically using the method described by Moench (1993). The inclusion of an empirical constant to partially account for effects of delayed yield is found to improve the match between theoretical type curves and field data. Type-curve matching can be used to determine the usual hydraulic parameters as well as the empirical constant.

Boundary-Value Problem

Neuman (1972, 1974) obtained a solution to the boundary-value problem for flow to fully and partially penetrating wells in a compressible, unconfined aquifer. The governing equation is

$$\frac{\partial^2 h}{\partial r^2} + \frac{1}{r}\frac{\partial h}{\partial r} + \frac{K_z}{K_r}\frac{\partial^2 h}{\partial z^2} = \frac{S}{bK_r}\frac{\partial h}{\partial t} \tag{1}$$

The symbols are defined in the Notation. The initial and boundary conditions are

$$h(r, z, 0) = 0 \tag{2}$$

$$h(\infty, z, t) = 0 \tag{3}$$

$$\frac{\partial h}{\partial z}(r, 0, t) = 0 \tag{4}$$

$$\lim_{r \to 0} \int_{b-l}^{b-d} r \frac{\partial h}{\partial r} dz = \frac{Q}{2\pi K_r} \tag{5}$$

$$K_z \frac{\partial h}{\partial z}(r, b, t) = -S_y \frac{\partial h(r, b, t)}{\partial t} \tag{6}$$

To be complete, equation (5) should also state that flow in the radial direction along the axis of the pumped well above and below the screened section is zero. Equation (6) is the condition that permits the water table to move freely in the vertical direction. With regard to the zone above the water table where water is held under tension, equation (6) implies that the equilibrium profile of soil moisture versus depth, in the unsaturated and nearly saturated zone, moves instantaneously in the vertical direction by an amount equal to the change in altitude of the water table.

Boulton (1954) defined the rate of drainage per unit area from fine-grained material overlying the aquifer at time t as

$$q(r, b, t) = \alpha_1 S_y \int_0^t \frac{\partial h}{\partial t'} e^{-\alpha_1(t-t')} dt' \quad (7)$$

where α_1 is an empirical constant. Boulton's α is subscripted in this paper to avoid confusion because it is used differently and takes on a slightly different meaning. Applying Darcy's law and assuming one-dimensional vertical flow in the zone above the water table, equation (7) becomes

$$K_z \frac{\partial h}{\partial z}(r, b, t) = -\alpha_1 S_y \int_0^t \frac{\partial h}{\partial t'} e^{-\alpha_1(t-t')} dt' \quad (8)$$

As discussed in the introduction, (8) can be derived by assuming vertical flow in a column of porous material above the water table under the conditions: (1) that the water and porous material are incompressible, and (2) that drainage from the unsaturated part of the column occurs instantaneously. Cooley and Case (1973) applied the equation as a free-surface condition to obtain an analytical solution to the problem of flow to a well that fully penetrates an aquifer overlain by a water-table aquitard. They found that the solution compared favorably with a numerical model developed by Cooley (1971) that made none of the linearizations required in the development of the analytical solution. Cooley (1971) also showed that Boulton's convolution integral is an adequate approximation for the free-surface condition in both a water-table aquifer and an aquifer with an overlying water-table aquitard.

The theoretical basis for use of (8) at the water table is weak. In this paper the equation is assumed to be nothing more than an empirical relation approximating the vertical flux of water at the base of the capillary fringe that occurs in response to temporally varying hydraulic head at the water table. The hydraulic properties of the porous material above and below the water table are assumed to be identical, in accordance with the assumption of aquifer homogeneity. No particular physical meaning is attached to the empirical constant α_1 other than that its inverse may be thought of as a relaxation coefficient, or a characteristic time scale. Boulton (1963) defined $1/\alpha$ as the "delay index." Because the convolution relation is used differently here than by Boulton, evaluation of the parameter by use of Boulton's type curves may not yield the same value as evaluation by the theory proposed in this paper.

Substituting the dimensionless parameters listed in Table 1 into equations (1)-(5) and (8), and assuming uniform flux along the screened section of the wellbore so that the integral in (5) drops out, one obtains the following boundary-value problem:

$$\frac{\partial^2 h_D}{\partial r_D^2} + \frac{1}{r_D}\frac{\partial h_D}{\partial r_D} + K_D \frac{\partial^2 h_D}{\partial z_D^2} = \frac{1}{r_D^2}\frac{\partial h_D}{\partial t_D} \quad (9)$$

$$h_D(r_D, z_D, 0) = 0 \quad (10)$$

$$h_D(\infty, z_D, t_D) = 0 \quad (11)$$

$$\frac{\partial h_D}{\partial z_D}(r_D, 0, t_D) = 0 \quad (12)$$

$$\lim_{r_D \to 0} r_D \frac{\partial h_D}{\partial r_D} = -\frac{2}{(\ell_D - d_D)} \quad (13)$$

$$\frac{\partial h_D}{\partial z_D}(r_D, 1, t_D) = -\gamma \int_0^{t_D} \frac{\partial h_D}{\partial \tau} e^{-\gamma\sigma\beta(t_D-\tau)} d\tau \quad (14)$$

The dimensionless form of (6), which (14) replaces, is

$$\frac{\partial h_D}{\partial z_D}(r_D, 1, t_D) = -\frac{1}{\sigma\beta}\frac{\partial h_D(r_D, 1, t_D)}{\partial t_D} \quad (14')$$

It can be seen that with the introduction of the dimensionless time, t_D, there results a natural dimensionless parameter, γ, composed of the empirical constant α_1, vertical hydraulic conductivity, saturated thickness, and specific yield. This parameter is the inverse of a dimensionless expression given by Boulton (1970, Appendix 3). In another paper, Boulton and Pontin (1971) introduce dimensionless parameters composed of the same quantities but with empirical constants that take on different physical meanings.

Application of the method of Laplace transformation to (9)-(14) leads to the following subsidiary equations:

$$\frac{d^2\bar{h}_D}{dr_D^2} + \frac{1}{r_D}\frac{d\bar{h}_D}{dr_D} + K_D \frac{d^2\bar{h}_D}{dz_D^2} = \frac{p\bar{h}_D}{r_D^2} \quad (15)$$

$$\bar{h}_D(\infty, z_D, p) = 0 \quad (16)$$

$$\frac{d\bar{h}_D}{dz_D}(r_D, 0, p) = 0 \quad (17)$$

$$\lim_{r_D \to 0} r_D \frac{d\bar{h}_D}{dr_D} = \frac{-2}{p(\ell_D - d_D)} \quad (18)$$

$$\frac{d\bar{h}_D}{dz_D}(r_D, 1, p) = -\frac{\bar{h}_D p}{\sigma\beta + p/\gamma} \quad (19)$$

For comparison the Laplace transform of (14′) is

$$\frac{d\bar{h}_D}{dz_D}(r_D, 1, p) = -\frac{\bar{h}_D p}{\sigma\beta} \quad (19')$$

The Laplace transform variable, p, is inversely related to t_D.

The solution to Neuman's boundary-value problem defined by (15)-(19) and (19′) is given by Moench (1993).

Table 1. Dimensionless Expressions

t_D	Tt/r^2S
t_{Dy}	Tt/r^2S_y
h_D	$4\pi T(h_i - h)/Q$
r_D	r/b
z_D	z/b
K_D	K_z/K_r
ℓ_D	ℓ/b
d_D	d/b
β	$K_z r^2/(K_r b^2)$
σ	S/S_y
γ	$\alpha_1 b S_y/K_z$

Comparison of (19) with (19′) reveals that the Laplace transform solution to the present problem can be written down directly from Moench (1993) by replacing $\sigma\beta$ by $\sigma\beta + p/\gamma$. Thus, the new solution is written as:

$$\bar{h}_D(\gamma, \sigma, \beta, z_D, p) = \bar{h}_{DT} + \Delta\bar{h}_{DH} + \Delta\bar{h}_{DN} \qquad (20)$$

where

$$\bar{h}_{DT} = \frac{2\, K_0(\sqrt{p})}{p}$$

$$\Delta\bar{h}_{DH} = \frac{2}{\pi(\ell_D - d_D)} \sum_{n=1}^{\infty} \frac{1}{n} \cos[n\pi(1 - z_D)]$$

$$\cdot\, [\sin(n\pi\ell_D) - \sin(n\pi d_D)]\, \frac{2\, K_0(\sqrt{\mu})}{p}$$

$$\Delta\bar{h}_{DN} = \frac{2}{\beta} \int_0^{\infty} \bar{F}(p, y)\, dy$$

$$\bar{F}(p, y) = \frac{\delta \cosh(\Psi z_D)\, y\, J_0(y)}{\Psi^2 \sinh(\Psi)\, [\Psi(\sigma\beta + p/\gamma) + p \coth(\Psi)]}$$

$$\delta = \frac{\sinh[\Psi(1 - \ell_D) - \sinh[\Psi(1 - d_D)]}{(\ell_D - d_D)\sinh(\Psi)}$$

$$\Psi^2 = (y^2 + p)/\beta$$

$$\mu = p + n^2\pi^2\beta$$

K_0 is the modified Bessel function of the second kind and order zero, and J_0 is the Bessel function of the first kind and order zero.

Because p is inversely related to t_D, the term p/γ, in the expression for $\bar{F}(p, y)$, becomes small relative to the constant product $\sigma\beta$ as either γ or dimensionless time increases and consequently (20) reduces to the Neuman solution. Equation (20) is evaluated in the manner described by Moench (1993) with no added complications. A revised version of the computer code (WTAQ1) is available from the author upon request.

Theoretical Responses

It is of interest to compare hypothetical responses (dimensionless drawdown, h_D, versus dimensionless time, t_{Dy}) produced by the Neuman model with the responses obtained by equation (20) for various values of γ in piezometers located at different vertical and horizontal positions in the aquifer. Figures 1 and 2 illustrate responses that occur as a result of holding β and σ constant while varying γ, or holding γ and σ constant while varying β. Since γ contains parameters that are shared by both β and σ, it is perhaps easiest to envision what is going on in Figures 1 and 2 by assuming that γ varies only with α_1 and that β varies only with r. Thus, Figure 1 illustrates the theoretical drawdown that occurs at a fixed distance, r, as α_1 is varied and Figure 2 illustrates the theoretical drawdown for two fixed values of α_1 as the distance, r, is varied. It should be noted that σ is held constant in these illustrations.

For a fully penetrating pumped well and fixed values of σ and β, Figure 1a shows the response at the top of the

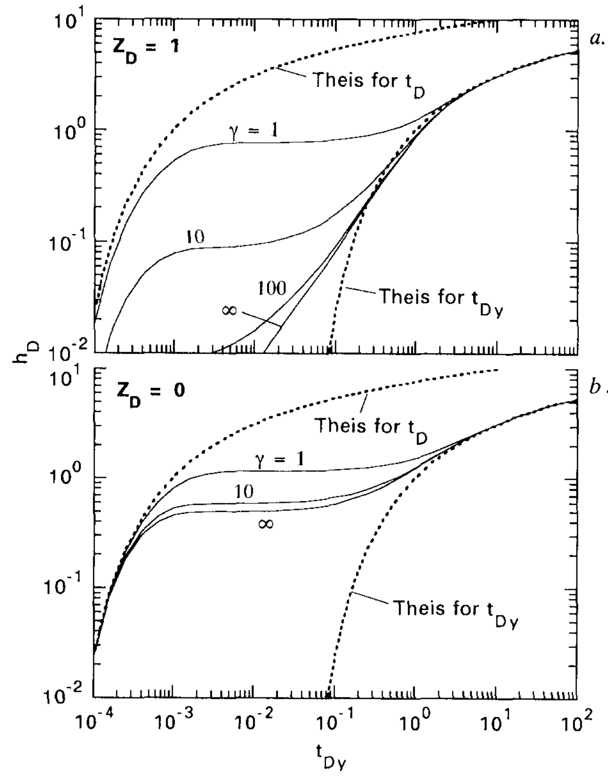

Fig. 1. Theoretical responses for various values of γ compared with the response for instantaneous drainage ($\gamma = \infty$): (a) for a water-table piezometer ($z_D = 1$); (b) for a piezometer located at the bottom of the aquifer ($z_D = 0$). [$\beta = 1$ and $\sigma = 10^{-3}$; $\ell_D = 1$ and $d_D = 0$].

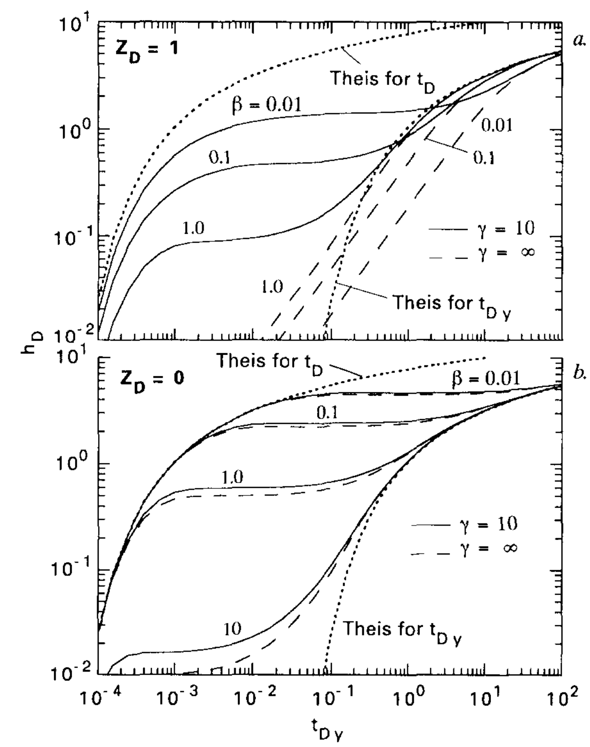

Fig. 2. Theoretical responses for various values of β with $\gamma = 10$ compared with the response for instantaneous drainage ($\gamma = \infty$): (a) for a water-table piezometer ($z_D = 1$); (b) for a piezometer located at the bottom of the aquifer ($z_D = 0$). [$\sigma = 10^{-3}$; $\ell_D = 1$ and $d_D = 0$].

aquifer ($z_D = 1$), and Figure 1b shows the response at the bottom of the aquifer ($z_D = 0$) for various values of γ. For small values of α_1 (large relaxation time) water is slow to drain from the unsaturated zone and the theoretical drawdown is enhanced. As α_1 (or γ) becomes large the aquifer response approaches the response obtained by the Neuman model ($\gamma = \infty$) for instantaneous release of water from the unsaturated zone. Comparison of Figures 1a and 1b demonstrates that the effect of delayed drainage is more apparent at the water table than at the bottom of the aquifer. This difference provides a basis for estimating γ by trial and error type-curve matching.

For a fully penetrating pumped well and fixed values of σ and γ, Figure 2a shows responses at the top of the aquifer and Figure 2b shows responses at the bottom of the aquifer for various values of β. The responses for $\gamma = \infty$ (actually, $\gamma = 10^6$ in the computations) are identical to the solution obtained by the Neuman model. As dimensionless time increases the responses for $\gamma = 10$ asymptotically approach the responses obtained by the Neuman model. Comparison of Figures 2a and 2b again demonstrates that the effect of delayed drainage manifests itself to a greater extent in water-table piezometers than in piezometers located at depth in the saturated zone. Figure 2a also shows that enhanced drawdown due to delayed drainage is most apparent near the pumped well (small β).

Applications to Aquifer-Test Analysis

Type-curve analyses of pumping tests that were conducted in two relatively homogeneous, glacial outwash, water-table aquifers with significantly different hydraulic conductivities (see Table 3) were carried out by Moench (1994). The aquifer tests referred to are the Borden site pumping test described by Nwankwor et al. (1984) and the Cape Cod site pumping test described by Moench et al. (1995). The reader is directed to the cited literature for particulars of the tests not covered in this paper. The analyses were based on the Neuman (1974) model for flow to a partially penetrating well that is pumped at a constant rate. In each instance a single "global" match point on composite (t/r^2) plots was obtained for nearly all the available drawdown data. The data were found to agree reasonably well with theoretical responses and unambiguous estimates of the aquifers' hydraulic properties were obtained. The early and intermediate time drawdown in piezometers located near the water table departed significantly, however, from the theoretical type curves, and it was not possible to obtain diagnostic estimates of vertical hydraulic conductivity based on use of the water-table piezometers alone. For reasons discussed by Moench (1994), no attempt was made to evaluate aquifer storativity at either site.

Cape Cod. At the Cape Cod site the water table at the time of the test was located at a depth of approximately 20 feet below land surface and the initial saturated thickness was estimated to be 160 feet. The pumped well was screened from 13.1 to 60 feet below the water table ($\ell_D = 0.375$ and $d_D = 0.082$) and was pumped at a constant rate of 320 gpm for 72 hours. The drawdown measured in three of the piezometers located near the water table are used here. The positions

Table 2. Positions of the Piezometers for the Cape Cod Site[a]

Piezometer number	r (ft)	Depth[b] (ft)	r_D^2	z_D
F505-032	23.9	10.7	0.022	0.93
F504-032	46.6	9.6	0.085	0.93
F383-032	93.0	12.1	0.34	0.92

[a] Dimensionless ratios, r_D and z_D, based on a saturated thickness of 160 feet.
[b] Approximate depth below the initial water table to the top of the two-foot long piezometer screen.

of these piezometers are given in Table 2. Figure 3a shows a composite plot of measured drawdown with type curves, which had been obtained by Moench et al. (1995) for the specified piezometers, superimposed. A Theis type curve for t_{Dy} is also shown for reference purposes. The match point for $h_D = 1$ and $t_{Dy} = 1$ is indicated in the figure ($h_i - h = 0.09$ ft and $t/r^2 = 6 \times 10^{-3}$ min/ft^2). This same match point was found for all the piezometers and observation wells analyzed by Moench et al. (1995). See Table 3 for values of the parameters obtained. It is clear from inspection of Figure 3a that the late-time match between measured and theoretical drawdown is reasonably good but the measured drawdown at early and intermediate time is significantly underestimated. It was reasoned by Moench et al. (1995) that these observations could be partially explained by the mechanism of delayed drainage discussed by Nwankwor et al. (1992).

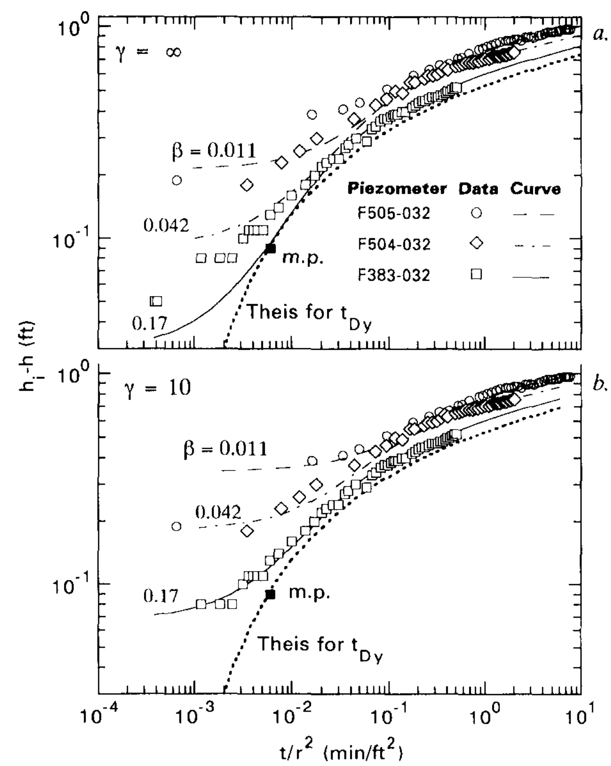

Fig. 3. Composite plot of drawdown at the Cape Cod site and comparison with type curves for the indicated piezometers located near the water table: (a) assuming instantaneous drainage ($\gamma = \infty$), (b) assuming noninstantaneous drainage ($\gamma = 10$). The indicated match point (m.p.) is for $t_{Dy} = 1$ and $h_D = 1$. [$K_z/K_r = 0.5$; $\sigma < 10^{-3}$].

Table 3. Summary of the Hydraulic Parameters Evaluated

Name of the site	T *(m^2/s)*	b *(m)*	K_z/K_r	K_z *(m/s)*	K_r *(m/s)*	S_y	α_1 (hr^{-1})
Borden	8.0×10^{-4}	6.7	0.3	3.6×10^{-5}	1.2×10^{-4}	0.24	0.24
Cape Cod	5.9×10^{-2}	48.8	0.5	6.0×10^{-4}	1.2×10^{-3}	0.23	1.96

Figure 3b shows that the match at early and intermediate time can be improved by use of the solution proposed in this paper. Holding K_z/K_r (and hence the values of β) equal to 0.50, the value obtained by Moench et al. (1995), the improved match was obtained by varying γ and comparing the resulting type curves with the drawdown data until a "best" match was obtained by visual inspection with the result that $\gamma = 10$. The ratio K_z/K_r was held constant in order not to change the fit to the drawdown in the piezometers located at depth and to the late-time drawdown in piezometers located near the water table.

Borden. At the Borden site, the water table at the time of the test was located at a depth of about 2.3 m below land surface and the initial saturated thickness was 6.7 m. The pumping well screen occupied the lower 60 percent of the aquifer ($\ell_D = 1.0$, $d_D = 0.4$) and was pumped at the constant rate of 1.0 liter/s for 64 hours. The drawdown measured in two of the (0.35 m long) observation piezometers, P1 and WS2, each located at a radial distance of 5 m from the pumped well and at depths of 5.0 m and 0.3 m, respectively, is used in this analysis. Figure 4 is a composite plot of the measured drawdown with two sets of type curves superimposed. One set includes the two type curves obtained by Moench (1994) using the Neuman model ($\gamma = \infty$), and the other set includes the modified type curves obtained by varying γ to improve the fit to the early and intermediate time drawdown data. The Theis curve for t_{Dy} is shown for reference purposes. The match point for $h_D = 1$ and $t_{Dy} = 1$ ($h_i - h = 0.1$ m and $t/r^2 = 5.0$ min/m^2) and value of the ratio K_z/K_r are chosen to coincide with the match point obtained by Moench (1994) and, as with the analysis of the Cape Cod site data, is based on a visual fit of the Neuman (1974) type curves to nearly all of the available data. The best visual match of the drawdown in these two piezometers appears to occur with $\gamma = 3$. The ratio K_z/K_r was held equal to 0.3, the value obtained by Moench et al. (1995), in order not to alter the aquifer parameters obtained previously.

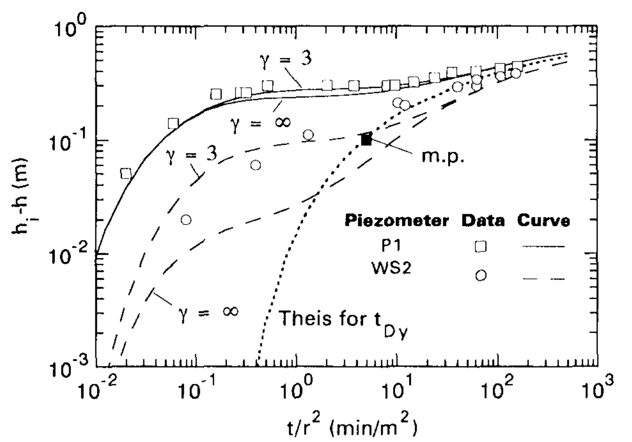

Fig. 4. Composite plot of drawdown at the Borden site for one deep piezometer (P1, where $z_D = 0.25$) and one water-table piezometer (WS2, where $z_D = 0.95$) compared with type curves assuming instantaneous drainage ($\gamma = \infty$), and noninstantaneous drainage ($\gamma = 3$). The indicated match point (m.p.) is for $t_{Dy} = 1$ and $h_D = 1$. [$K_z/K_r = 0.3$; $\beta = 0.168$; $\sigma = 1.5 \times 10^{-2}$].

Although the theoretical response for piezometer WS2 using $\gamma = 3$ is a significant improvement over the theoretical response for $\gamma = \infty$, the theoretical response does not compare nearly as well as the responses shown in Figure 3b for the piezometers located near the water table at the Cape Cod site. This observation suggests that the drainage process is less well-represented by a single-parameter exponential relation in the relatively low hydraulic conductivity material at the Borden site than in the higher hydraulic conductivity material at the Cape Cod site.

The hydraulic properties obtained for these two aquifer tests by Moench (1994) and the empirical constant α_1, obtained in this paper are summarized in Table 3. The order of magnitude difference in relaxation time ($1/\alpha_1$) for the two tests is consistent with the order of magnitude difference in vertical hydraulic conductivity.

With the aid of laboratory column experiments to obtain drainage-versus-time relations, similar to those of Vachaud (1968), performed on representative samples of aquifer material taken from near the water table, it may be possible to make an estimate of the empirical constant α_1 by fitting, however crudely, a single-parameter exponential relation to the discharge data. Such an independent estimate of α_1 might be used to remove some of the uncertainty in the evaluation of hydraulic parameters by type-curve analysis, particularly if deep-seated piezometers are unavailable and one must rely solely on drawdown measured in water-table piezometers.

Summary and Conclusion

Boulton's convolution integral is used as a free-surface condition in the three-dimensional, axisymmetric boundary-value problem for flow to a partially penetrating well in a water-table aquifer. This is done in order to approximately represent the gradual release of water from the zone above the water table. A Laplace transform solution to the modified boundary-value problem is obtained and a dimensionless fitting parameter, γ, is introduced. This parameter can be evaluated in addition to the parameters generally obtained by the method of type-curve analysis. The proposed solution allows for a better match of theoretical response and measured drawdown. It reduces to Neuman's solution as time increases or as the rate of release of water (as controlled by α_1) from the zone above the water table increases. Although the drainage process is not well-represented by an exponential decay in time, the use of Boulton's concept of delayed yield in the analytical treatment of flow to a well in a water-table aquifer is an improvement over the assumption of instantaneous drainage. It is recommended that pumping tests conducted in water-table aquifers be carried out using piezometers located at various depths and distances from the pumped well. If the aquifer is homogeneous, it should be possible to use water-table piezometers as well as deep-seated piezometers and obtain a consistent set of aquifer parameters.

Notation

α_1 empirical constant, T^{-1};

b initial saturated thickness of aquifer, L;

d vertical distance from initial water table to top of pumped well screen, L;

h hydraulic head, L;

h_i initial hydraulic head, L;

K_z hydraulic conductivity in the vertical direction, LT^{-1};

K_r hydraulic conductivity in the horizontal direction, LT^{-1};

ℓ vertical distance from initial water table to bottom of pumped-well screen, L;

Q pumping rate, L^3T;

q rate of drainage per unit area, LT^{-1};

r radial distance from axis of pumping well, L;

S storativity;

S_y specific yield;

T transmissivity, L^2T^{-1};

t time since start of pumping, T; and

z vertical distance above bottom of aquifer, L.

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